Aliah University

End-Semester Examination (Autum Semester) - 2021

(For 3rd Year BTECH)

Paper Name: Design and Analysis of Algorithm **Full Marks: 80**

Paper Code: CSE305 **Time: 3 hrs**

(Answer any 8 questions) **10 × 8 = 80**

1. Discuss the main steps of divide and conquer technique for solving problem. In light of this explain the quick sort algorithm. Analyze time complexity of quick sort algorithm (write the recurrence relation and solve it). When it achieves quadratic complexity.

   2+ 2+4+2=10

2. Given an array of size n devise an algorithm for finding minimum and second minimum from the array using divide and conquer technique. Analyze the complexity of your algorithm.

   5+ 5 =10

3. Define big-O, big-theta notation. Write the graphical interpretation of it. In which complexity class the following function belong? $f(n) = 5n^3 + 6n$. When the quick sort has best case time complexity? 3+2 +3+2=10

4. Write and explain the recurrence relation for solving matrix chain multiplication problem using dynamic programming technique. solve the following matrix chain multiplication problem using dynamic programming:

   A1:20x5, A2: 5x50, A3:50x10, A4: 10x30

   3 + 7=10

5. Consider two sorted array A=[3 6 19 30 55 60] and B=[15 18 20 35 40 44]. How you merge these two using merging technique. What will be the number of comparisons. Proof that any comparison-based sorting technique has the worst-case time complexity O(nlogn)

   4+3 +3=10

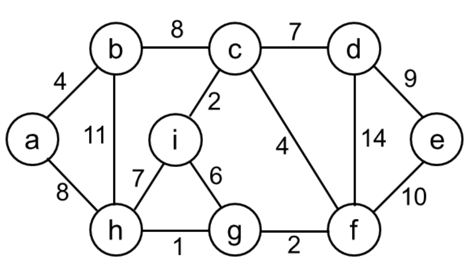
6. State the 0/1 Knapsack problem. Consider instance of the 0/1 knapsack problem as below with P depicting the value and w depicting the weight of each item whereas M denotes the total weight carrying capacity of the knapsack. Find optimal answer using dynamic programming technique.

   P: [40 10 50 30 60] W = [80 10 40 20 90] M : 110

   If fractional part of items is allowed to take then what will be the optimal solution.

   2+5+3=10

7. Define minimum spanning tree of a graph. What is cut-edge? Find the minimum spanning tree of the following graph using Prim's algorithm. 2 +2 +6=10



8. What is max heap? Describe the Max_Heapify procedure (write the pseudo code). From the following array build a Max Heap (draw step by step diagram for building the Max Heap).

4	1	3	2	16	9	10	14	8	7

2+4+4=10

9. What is backtracking search. Define N-queens problem. Draw a backtrack search tree for 4-queens problem. What is difference between divide and conquer and dynamic programming technique? 2+ 2+4+2=10

10. Describe depth first search and breadth first search technique with an example. Write an algorithm for depth first search. When a decision problem is called NP complete? Give an example of NP complete problem. 3+ 4+2+1=10